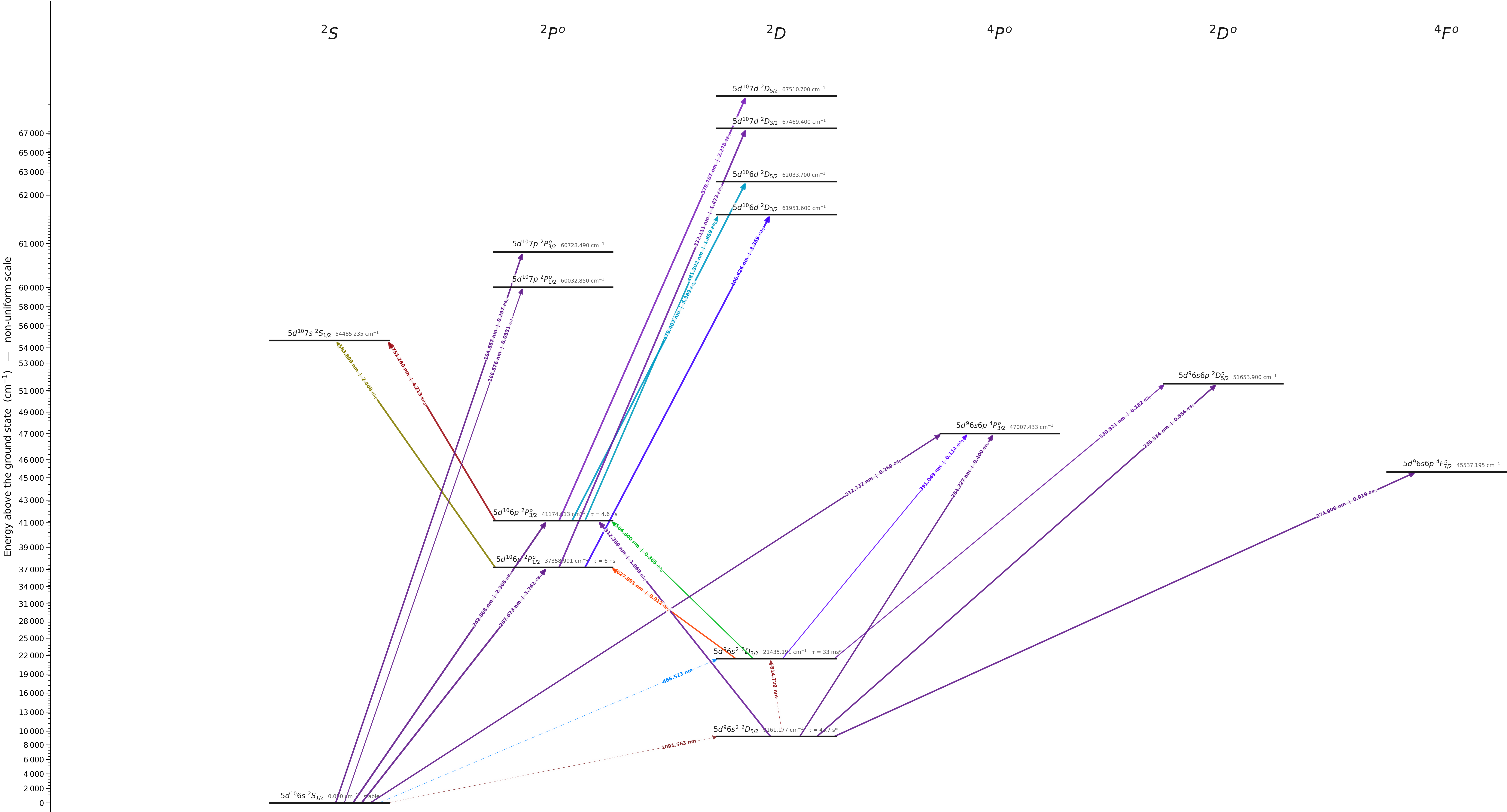
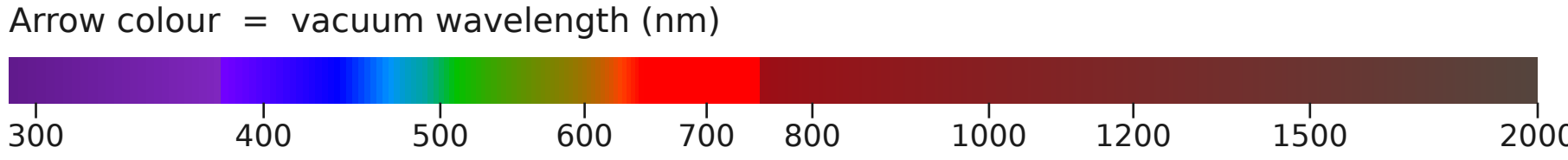


Gold

Au (Au I). Strongest classified transitions below 2 μm (NIST): 15 levels, 23 lines.

Au⁺ ionisation limit 74409.11 cm⁻¹ = 9.22555 eV (134.392 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U\|er\|U'\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U\ er\ U'\rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d ⁹ 6s ² 2D _{5/2} – 5d ⁹ 6s ² 2D _{3/2}	814.7294	814.5054	367.9657	–	2.990e+01	4.823e-06	98.7	main decay of 2D3/2
5d ¹⁰ 6s 2S _{1/2} – 5d ⁹ 6s ² 2D _{5/2}	1091.5628	1091.2639	274.6452	–	2.290e-02	3.642e-09	100	E2 clock transition proposed for a Au lattice clock

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁹⁷Au hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5d ¹⁰ 6s 2S _{1/2}	3049.660	–	F=1: -3812.07 F=2: +2287.24
5d ⁹ 6s ² 2D _{5/2}	80.236(3)	-1049.781(11)	F=1: -1156.09 F=2: -155.79 F=3: +557.32 F=4: +38.44
5d ⁹ 6s ² 2D _{3/2}	199.84250(20)	-911.0766(5)	F=0: -1888.26 F=1: -777.34 F=2: +533.43 F=3: +221.88
5d ¹⁰ 6p 2P _{3/2} ^o	14.0(5)	327.6(16)	F=0: +357.00 F=1: +43.40 F=2: -256.20 F=3: +113.40

Measured A, B constants; sources in the data file.